

FasiriNET

Software Quality & Usability Test Plan

Version 1.0 · Aligned with ISO/IEC 25010 & WCAG 2.1 AA

This document defines the complete testing strategy used to demonstrate that FasiriNET — an African-languages voice-dataset platform — meets professional software-quality standards. It pairs the platform's built-in automated test runner (accessible at **/admin** → **Tests**) with the manual evaluation procedures that require human judgement, external tooling, or real-user participation.

Quality Model

Tests are organised around the eight quality characteristics defined in ISO/IEC 25010: **Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability**.

Quality Characteristic	Coverage	Method
Functional Suitability	Automated	UX Flow Contracts, RPCs, Business Rules
Performance Efficiency	Automated	Performance & Latency suite (p95 budgets)
Compatibility	Automated	Browser/Device Capability suite
Usability	Mixed	Accessibility suite + moderated SUS sessions
Reliability	Automated	Error Resilience + Data Integrity suites
Security	Mixed	RLS suite + manual OWASP penetration test
Maintainability	Manual	Code review, ESLint, TypeScript strict
Portability	Mixed	Compatibility suite + cross-browser matrix

1. Automated Test Suites

All suites below run from **Admin Panel** → **Tests**. Click **Run all tests** to execute them against the live production backend. Each test reports pass/fail, duration, and details.

#	Suite	What it proves
1	Authentication & Roles	Verifies sign-in works, role assignments are correct, and unauthorised users cannot escalate.
2	RLS & Data Access	Confirms Row-Level Security blocks anonymous reads on sensitive tables (profiles, voice_responses, transcriptions, user_roles).
3	RPC Functions	Calls each Postgres function (has_role, get_question_counts, claim_random_transcription, release_transcription_lock) with valid inputs.
4	Storage	Validates the voice-recordings bucket is reachable, signed URLs work, and uploaded objects are listable.
5	Edge Functions	Probes CORS preflight on reset-password, migrate-drive-audio, and upload-to-gcs endpoints.
6	Data Integrity	Detects orphaned rows, stale transcription locks, and broken foreign-key references.
7	Business Rules	Asserts the 30s recording cap, 3-unique-responses retirement rule, and 3-edits-max transcription rule.
8	Performance & Latency	Times key queries against published budgets: counts < 1.5s, list fetches < 1s, signed URL < 800ms, profile read < 500ms.
9	Accessibility (a11y)	WCAG 2.1 AA smoke checks: alt text, button names, input labels, heading hierarchy, contrast ratio $\geq 4.5:1$, lang attribute, viewport meta.
10	Browser & Device Compatibility	Confirms MediaRecorder, getUserMedia, IndexedDB, Service Worker, localStorage, Web Audio, Fetch, and online-status APIs are present.
11	User Journey Contracts	End-to-end read-only checks behind signup, recording, transcription, and dashboard flows.
12	Error Resilience	Forces error paths (bad credentials, missing records, anon RPC, missing storage object, unknown edge fn, invalid UUID) and asserts graceful typed errors.

2. Manual Test Procedures

These categories require humans, external tooling, or real participants. They cannot be executed from the in-app dashboard but are essential evidence of usability.

2.1 Moderated Usability Sessions (SUS)

Goal

Measure perceived usability with the System Usability Scale (Brooke, 1996). Target SUS score ≥ 75 (good).

Protocol

- Recruit 8–12 native speakers per dialect (Kinyarwanda, Swahili, Lingala, etc.).
- 30-minute moderated session per participant (in-person or video call).
- Ask participant to think aloud while completing tasks. Record screen + audio.
- Administer the 10-question SUS questionnaire after the tasks.
- Capture: time-on-task, task success rate, error count, qualitative feedback.

Task list

- **T1** — Sign up as a Recording Volunteer and complete profile (target ≤ 3 min).
- **T2** — Record three voice responses to questions on the dashboard.
- **T3** — Use the Skip / Re-record flow on at least one question.
- **T4** — Find your past responses on the "My Responses" page.
- **T5** — Sign up as a Transcriber, wait for approval simulation, then transcribe one audio.
- **T6** — Recover from an offline state mid-recording (toggle airplane mode).

Pass criteria

- $\geq 90\%$ of participants complete all six tasks unaided.
- Mean SUS score ≥ 75 .
- Zero participants encounter a blocking error (crash, frozen state, data loss).

2.2 Load & Stress Testing (k6)

Validate the platform behaves under realistic concurrent load. Tooling: **Grafana k6** (open source).
Target environment: production replica.

Scenarios

- **Steady load** — 50 virtual users browsing dashboard for 10 minutes.
- **Recording spike** — 100 concurrent volunteers uploading a 30s audio file each over 2 min.
- **Transcription burst** — 40 transcribers claiming + submitting transcripts simultaneously.
- **Soak test** — 20 VUs sustained for 1 hour to detect memory leaks or lock buildup.

Target metrics

Metric	Target	Failure Threshold
p95 API latency	< 1.5 s	> 3 s
Error rate	< 0.5 %	> 2 %
Audio upload success	≥ 99 %	< 95 %
Throughput (req/s)	≥ 50	< 20
DB connection saturation	< 70 %	> 90 %

Sample k6 outline

```
import http from 'k6/http';
import { sleep, check } from 'k6';
export const options = { vus: 100, duration: '2m' };
export default function () {
  const res = http.post(URL + '/storage/v1/object/voice-recordings/x.webm', file, hdr);
  check(res, { 'upload 200': r => r.status === 200 });
  sleep(1);
}
```

2.3 Accessibility Audit (Manual)

The in-app a11y suite covers static checks. This section covers the dynamic and assistive-tech checks that require human verification.

Tools

- **axe DevTools** (Chrome extension) — full WCAG 2.1 AA scan on every public route.
- **Lighthouse** Accessibility audit — score ≥ 95 .
- **NVDA** (Windows) and **VoiceOver** (macOS/iOS) — screen-reader walkthrough.
- **Keyboard-only** — every flow completable without a mouse.

Checklist

- All interactive elements reachable via Tab in logical order.
- Focus indicators visible on every focusable element.
- Modals trap focus and restore on close.
- Voice recorder timer announced via aria-live region.
- Error toasts have role="alert" or aria-live="assertive".
- No information conveyed by colour alone.
- Touch targets $\geq 44 \times 44$ px on mobile.
- Page passes **axe DevTools** with zero violations on Login, Signup, Dashboard, Questions, Transcribe.

2.4 Cross-Browser / Device Matrix

Browser / OS	Versions	Devices
Chrome (desktop)	latest, latest-1	Windows 10/11, macOS
Safari (desktop)	latest, latest-1	macOS 13+
Firefox (desktop)	latest	Windows, macOS
Edge (desktop)	latest	Windows 11
Chrome (Android)	latest	Pixel, Samsung Galaxy
Safari (iOS)	iOS 16, 17, 18	iPhone 12+, iPad
Samsung Internet	latest	Galaxy S22+

Each row must complete tasks T1–T6 from §2.1 without functional regression.

2.5 Security Penetration Testing

OWASP Top 10 (2021) checklist mapped to the FasiriNET stack (React + Supabase + Edge Functions).

OWASP Risk	Mitigation in place	Manual verification
A01 Broken Access Control	Postgres RLS on every table; user_roles separation.	Attempt cross-user reads with a forged JWT.
A02 Cryptographic Failures	TLS 1.3, hashed passwords (Supabase Auth).	SSL Labs scan; verify HSTS header.
A03 Injection	Parameterised queries via supabase-js.	Fuzz inputs with sqlmap on public endpoints.
A04 Insecure Design	Threat model reviewed; least-privilege SECURITY DEFINER fns.	Architecture review.
A05 Security Misconfig	CSP headers, no debug routes in prod.	Headers scan (securityheaders.com).
A06 Vulnerable Components	Dependabot + npm audit weekly.	Run `npm audit --production`.
A07 Auth Failures	Email/password only, password strength meter, rate limiting.	Brute-force test on /login.
A08 Data Integrity Failures	Signed audio URLs; immutable transcription history.	Verify object ETags after upload.
A09 Logging Failures	Activity logs table + realtime presence.	Confirm log rows persist after admin actions.
A10 SSRF	Edge fns validate destination URLs.	Attempt to call internal IPs from upload-to-gcs.

3. Acceptance Criteria

The platform is considered **release-ready** when ALL of the following pass thresholds are met.

Category	Threshold	Source
Automated test pass rate	≥ 95 %	/admin → Tests
Critical-path test pass rate	100 %	Auth, RLS, UX Flow, Error Resilience suites
SUS score (mean)	≥ 75	Moderated usability sessions
Task success rate	≥ 90 %	Moderated usability sessions
p95 API latency under load	< 2 s	k6 recording-spike scenario
Audio upload success under load	≥ 99 %	k6 recording-spike scenario
Lighthouse Accessibility	≥ 95	Lighthouse on every public route
axe DevTools violations	0	Manual scan on all routes
OWASP Top-10 high-severity findings	0	Manual penetration test
npm audit high/critical	0	Dependency scan

4. Test Cadence

- **Every commit** — TypeScript compile, ESLint, dependency scan.
- **Every deploy** — Run all 12 automated suites from /admin → Tests.
- **Weekly** — Lighthouse + axe DevTools scan on production.
- **Monthly** — k6 load test against staging replica.
- **Quarterly** — Moderated usability sessions (8–12 users).
- **Bi-annually** — Full OWASP penetration test by external auditor.

5. Reporting

All automated runs export to JSON via the **Copy JSON** button in the Tests tab. Manual sessions are recorded in a shared evidence folder containing: video recordings, SUS scoresheets, k6 reports, axe scan results, and signed pen-test reports. The latest results are summarised on the public Quality & Trust page (planned).

— End of Test Plan v1.0 —